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IoT and Its Applications

TRAINING PARTNER

ENTHU TECHNOLOGY SOLUTIONS INDIA PVT LTD

10 / 12 / 2025 – 20 / 12 / 2025

SMART GREENHOUSE AUTOMATED SYSTEM USING ESP32

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Abstract

Agriculture is a major contributor to food security, and the use of greenhouse technology has become increasingly important for improving crop yield and quality. However, maintaining optimal environmental conditions inside a greenhouse through manual monitoring is labor-intensive, inefficient, and prone to human error. To overcome these challenges, this project presents the design and implementation of a **Smart Greenhouse Automated System using ESP32**, which ensures continuous monitoring and automatic control of critical environmental parameters required for healthy plant growth.

The proposed system employs a **DHT11 sensor** to measure temperature and humidity levels inside the greenhouse, a **soil moisture sensor** to determine the water content of the soil, and a **BH1750 light intensity sensor** to monitor the availability of ambient light. The ESP32 microcontroller acts as the central processing unit, collecting sensor data and making real-time control decisions based on predefined threshold values. When the temperature or humidity exceeds the optimal range, the system automatically activates an **exhaust fan** to regulate airflow and maintain a stable internal climate. If the soil moisture level drops below the required threshold, a **water pump** is switched ON to provide adequate irrigation and prevent plant stress. Similarly, when the light intensity falls below the desired level, an **LED grow light** is turned ON to ensure sufficient illumination for photosynthesis and plant development.

This automated approach significantly reduces the need for manual supervision while improving the accuracy and consistency of greenhouse management. The system ensures efficient utilization of water and energy resources, minimizes crop damage due to unfavorable environmental conditions, and enhances overall agricultural productivity. The use of ESP32 offers additional advantages such as low power consumption, high processing speed, and future scalability for wireless monitoring and IoT integration.

The Smart Greenhouse Automated System demonstrates how embedded systems and sensor-based automation can be effectively applied in modern agriculture. By providing a reliable, cost-effective, and efficient solution for environmental control, the proposed system supports sustainable farming practices and offers a practical foundation for advanced smart agriculture applications.

Introduction

Agriculture plays a crucial role in sustaining human life and supporting economic growth. With the increasing demand for food and the challenges posed by climate change, traditional farming methods often fail to provide consistent crop yield and quality. Greenhouse cultivation has emerged as an effective solution to overcome these challenges by offering a controlled environment for plant growth. However, maintaining optimal environmental conditions inside a greenhouse through manual monitoring is labor-intensive, time-consuming, and inefficient. This has led to the growing need for automated greenhouse systems that can continuously monitor and regulate environmental parameters with minimal human intervention.

A smart greenhouse system uses embedded systems and sensors to monitor key factors such as **temperature, humidity, soil moisture, and light intensity**, which directly influence plant growth and productivity. Proper control of these parameters ensures healthy plant development, efficient use of resources, and reduced crop loss. Automation not only improves accuracy and reliability but also helps farmers manage greenhouses more effectively, even with limited manpower.

In this project, a **Smart Greenhouse Automated System using ESP32** is developed to provide real-time monitoring and automatic control of greenhouse conditions. The ESP32 microcontroller is selected due to its high processing capability, low power consumption, and suitability for smart agriculture applications. The system integrates a **DHT11 sensor** to measure temperature and humidity, a **soil moisture sensor** to determine soil water content, and a **BH1750 light intensity sensor** to monitor ambient light levels inside the greenhouse.

Based on the sensor readings, the ESP32 controls various output devices to maintain optimal environmental conditions. When the temperature or humidity exceeds the predefined safe limits, an **exhaust fan** is automatically activated to regulate airflow and reduce excessive heat or moisture. If the soil moisture level falls below the required threshold, a **water pump** is switched ON to provide timely irrigation and prevent plant stress. Similarly, when light intensity is insufficient, an **LED grow light** is turned ON to ensure adequate illumination for photosynthesis and proper plant growth.

The proposed system significantly reduces the need for continuous manual supervision and minimizes human error. It also promotes efficient utilization of water and energy resources, making it both cost-effective and environmentally friendly. By automating critical greenhouse operations, this project demonstrates the practical application of embedded systems and sensor technology in modern agriculture. The Smart Greenhouse Automated System thus serves as a reliable and scalable solution that can be further enhanced with wireless communication and IoT features for remote monitoring and advanced agricultural management.

Problem Statement

Agriculture is highly dependent on environmental conditions such as temperature, humidity, soil moisture, and light intensity. In greenhouse cultivation, maintaining these parameters within optimal limits is essential for healthy plant growth and maximum crop yield. However, in many traditional greenhouse setups, monitoring and control of environmental conditions are carried out manually. This manual approach is time-consuming, inefficient, and often inaccurate, leading to inconsistent growing conditions and reduced productivity.

One of the major challenges in greenhouse management is the lack of continuous monitoring. Temperature and humidity can fluctuate rapidly due to external weather changes, and without timely regulation, plants may experience stress. Excessive temperature and humidity can cause plant diseases, reduced growth, and poor crop quality. In the absence of an automated system, exhaust fans are often operated manually, which may result in delayed response and improper climate control.

Another significant issue is improper irrigation management. Overwatering can lead to root damage and fungal infections, while underwatering can cause wilting and reduced plant growth. Conventional irrigation systems usually operate on fixed schedules without considering the actual soil moisture condition. This leads to wastage of water and inefficient use of resources. There is a need for a system that can automatically monitor soil moisture levels and control water supply accordingly

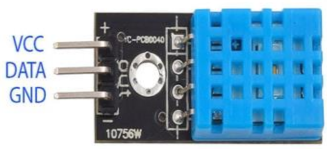
Light availability is another critical factor affecting photosynthesis and plant development. In many greenhouses, natural sunlight may be insufficient due to seasonal variations, cloudy weather, or structural limitations. Manual control of artificial lighting is inefficient and may not provide consistent light levels required for optimal plant growth. Hence, an automated lighting system that responds to real-time light intensity is required.

Additionally, the lack of integration between monitoring and control systems makes greenhouse management complex and labor-intensive. Small-scale and medium-scale farmers often cannot afford expensive commercial automation solutions, creating a demand for a low-cost, reliable, and efficient system.

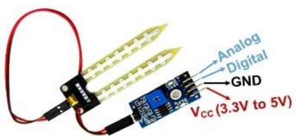
Therefore, the problem is to design and implement a Smart Greenhouse Automated System using ESP32 that can continuously monitor temperature, humidity, soil moisture, and light intensity using DHT11, soil moisture sensor, and BH1750. The system should automatically control an exhaust fan, water pump, and LED light based on real-time sensor data. Such a system aims to reduce manual intervention, optimize resource usage, maintain optimal greenhouse conditions, and improve crop productivity in a cost-effective manner.

Apparatus Required

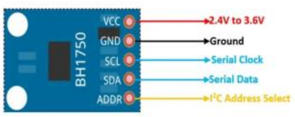
1. DHT11



2. Soil Moisture



3. BH1750

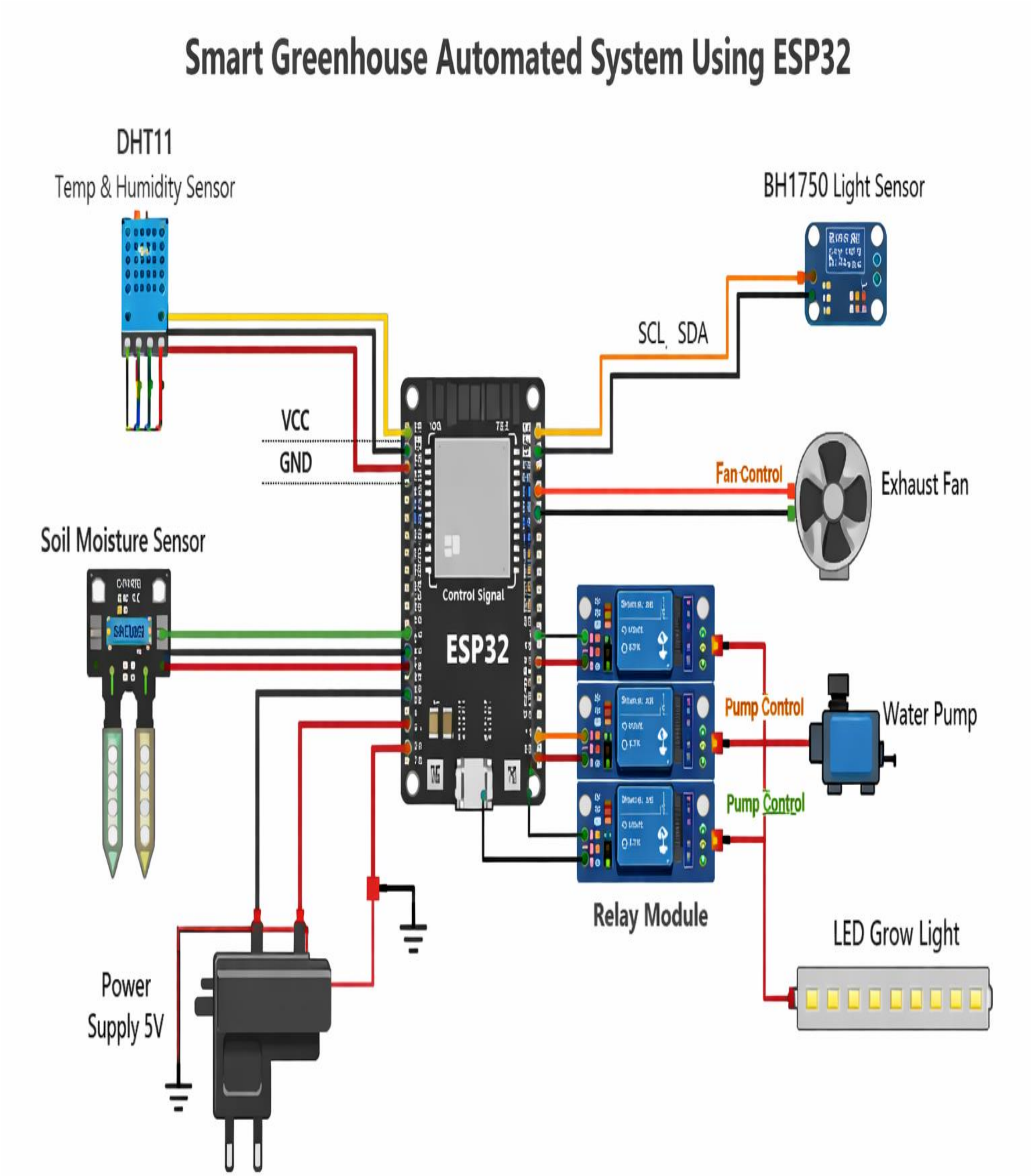


4. Exhaust FAN

5. Pump

6. LED
7. Relay

Circuit Diagram



Description

The Smart Greenhouse Automated System is designed to continuously monitor and regulate essential environmental parameters required for healthy plant growth using an ESP32 microcontroller. The system integrates multiple sensors and actuators to automate greenhouse operations, thereby reducing human intervention and improving efficiency. The primary parameters monitored in this system are temperature, humidity, soil moisture, and light intensity, which play a crucial role in plant development.

The ESP32 microcontroller acts as the central control unit of the system. It receives input data from all sensors, processes the information in real time, and controls the corresponding output devices based on predefined threshold values. The ESP32 is chosen due to its high processing speed, low power consumption, and compatibility with multiple sensors, making it suitable for smart agriculture applications.

The DHT11 sensor is used to measure temperature and humidity inside the greenhouse. These parameters directly affect plant health and growth rate. When the temperature or humidity exceeds the preset optimal range, the ESP32 activates an exhaust fan through a relay module. The exhaust fan helps in reducing excess heat and moisture by improving air circulation, thus maintaining a stable and suitable greenhouse environment.

A soil moisture sensor is used to monitor the water content present in the soil. Proper irrigation is essential for plant growth, and both overwatering and underwatering can harm plants. The soil moisture sensor continuously sends analog values to the ESP32. When the moisture level falls below the defined threshold, indicating dry soil conditions, the ESP32 automatically switches ON a water pump via a relay. The pump supplies water to the plants until sufficient moisture is restored, after which it is turned OFF automatically.

The BH1750 light intensity sensor is employed to measure ambient light levels within the greenhouse. Adequate light is essential for photosynthesis and overall plant development. The BH1750 communicates with the ESP32 using the I^2C protocol and provides accurate digital lux readings. When the measured light intensity is below the required level, the ESP32 turns ON an LED grow light to supplement natural light. Once sufficient light is detected, the LED is switched OFF to conserve energy.

All high-power devices such as the exhaust fan and water pump are interfaced using a relay module, ensuring electrical isolation and safe operation. The system operates continuously in a loop, constantly monitoring environmental conditions and responding automatically to any changes.

Overall, the Smart Greenhouse Automated System provides an efficient, reliable, and cost-effective solution for modern greenhouse management. By integrating sensor-based monitoring and automated control, the system enhances plant growth, optimizes resource usage, and demonstrates the practical application of embedded systems in smart agriculture.

Pseudo Code

```
#include <Wire.h>
#include <BH1750.h>
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT11
#define SOIL_PIN 34
#define FAN_PIN 16
#define PUMP_PIN 17
#define LIGHT_PIN 18

DHT dht(DHTPIN, DHTTYPE);
BH1750 lightMeter;

void setup() {
  Serial.begin(9600);

  pinMode(FAN_PIN, OUTPUT);
  pinMode(PUMP_PIN, OUTPUT);
  pinMode(LIGHT_PIN, OUTPUT);

  digitalWrite(FAN_PIN, LOW);
  digitalWrite(PUMP_PIN, LOW);
  digitalWrite(LIGHT_PIN, LOW);

  dht.begin();

  Wire.begin(21, 22);   // SDA, SCL
```

```
  lightMeter.begin();

  Serial.println("Smart Greenhouse System Started");
}

void loop() {

  /* -------- DHT11 -------- */
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  if (temp > 30) {
    digitalWrite(FAN_PIN, HIGH);   // FAN ON
  } else {
    digitalWrite(FAN_PIN, LOW);    // FAN OFF
  }

  /* -------- Soil Moisture -------- */
  int soilValue = analogRead(SOIL_PIN);

  if (soilValue < 2000) {
    digitalWrite(PUMP_PIN, HIGH);  // PUMP ON
  } else {
    digitalWrite(PUMP_PIN, LOW);   // PUMP OFF
  }

  /* -------- BH1750 -------- */
  float lux = lightMeter.readLightLevel();

  if (lux < 300) {
    digitalWrite(LIGHT_PIN, HIGH); // LIGHT ON
  } else {
    digitalWrite(LIGHT_PIN, LOW);  // LIGHT OFF
  }

  /* -------- Serial Monitor -------- */
  Serial.print("Temp: "); Serial.print(temp);
```

```
    Serial.print(" °C | Humidity: "); Serial.print(hum);
    Serial.print(" % | Soil: "); Serial.print(soilValue);
    Serial.print(" | Lux: "); Serial.println(lux);

    delay(2000);
}
```

Procedure

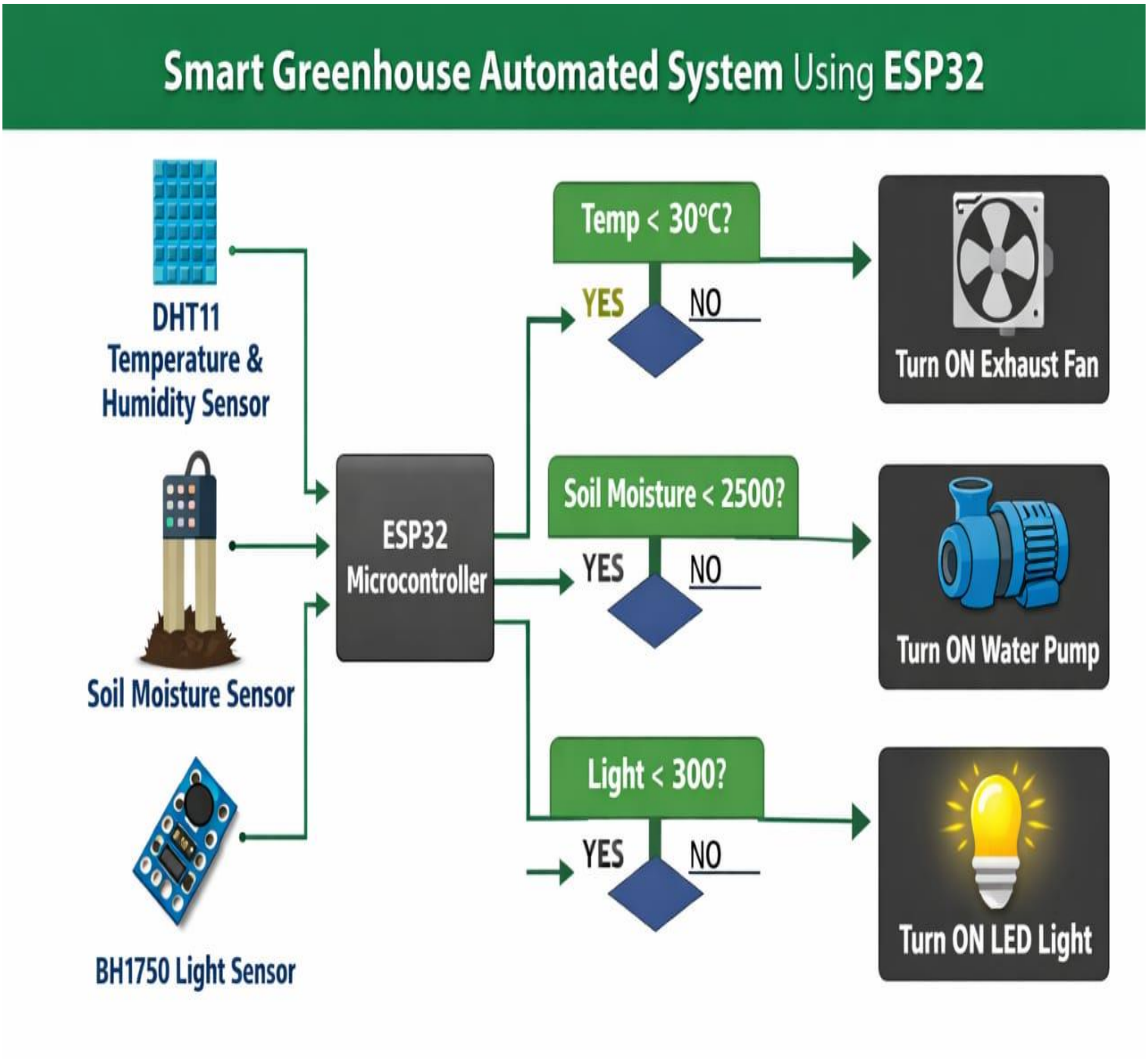
The procedure followed for designing and implementing the **Smart Greenhouse Automated System using ESP32** is described step by step as follows:

1. **Component Selection and Planning**
   The required components such as ESP32 microcontroller, DHT11 temperature and humidity sensor, BH1750 light intensity sensor, soil moisture sensor, relay module, exhaust fan, water pump, LED light, and power supply are selected. The operational thresholds for temperature, humidity, soil moisture, and light intensity are defined based on plant requirements.
2. **Hardware Setup and Circuit Assembly**
   The ESP32 is placed on a breadboard or mounting platform and powered using a regulated power supply. The DHT11 sensor is connected to a digital GPIO pin of the ESP32 for measuring temperature and humidity. The soil moisture sensor is connected to an analog input pin to read soil moisture levels. The BH1750 light sensor is interfaced with the ESP32 using I^2C communication (SDA and SCL pins). A relay module is used to connect the exhaust fan, water pump, and LED light to ensure safe switching of high-power devices.
3. **Software Development**
   The ESP32 is programmed using the Arduino IDE. Necessary libraries for DHT11 and BH1750 sensors are included. The program is written to read sensor data continuously, compare the readings with predefined threshold values, and control the output devices accordingly.
4. **Uploading the Program**
   The developed program is compiled and uploaded to the ESP32 using a USB cable. Any compilation or upload errors are corrected before proceeding further.
5. **System Initialization**
   Once powered ON, the ESP32 initializes all sensors and output devices.

Initial calibration of the soil moisture sensor is performed to ensure accurate readings.

6. **Sensor Data Acquisition**
   The ESP32 continuously reads temperature and humidity data from the DHT11 sensor, soil moisture levels from the soil moisture sensor, and light intensity values from the BH1750 sensor.
7. **Automatic Control Operation**
   If the temperature or humidity exceeds the set threshold, the ESP32 activates the relay to turn ON the exhaust fan. When the soil moisture level drops below the required value, the water pump is turned ON automatically to irrigate the plants. Similarly, if the light intensity measured by the BH1750 sensor is low, the ESP32 switches ON the LED light to provide sufficient illumination.
8. **Continuous Monitoring and Adjustment**
   The system operates continuously in a loop, adjusting the operation of the fan, pump, and LED in real time based on changing environmental conditions.
9. **Observation and Testing**
   The operation of the system is observed under different conditions to verify correct functioning. Sensor readings and actuator responses are checked to ensure reliable and accurate automation.

Flowchart



Smart Greenhouse Automated System Using ESP32
DHT11
Temperature &
Humidity Sensor
Soil Moisture Sensor
BH1750 Light Sensor
ESP32
Microcontroller
Temp < 30°C?
YES
NO
Soil Moisture < 2500?
YES
NO
Light < 300?
YES
NO
Turn ON Exhaust Fan
Turn ON Water Pump
Turn ON LED Light

Flowchart Explanation

The flowchart illustrates the working logic of the Smart Greenhouse Automated System using ESP32, which continuously monitors environmental parameters and controls greenhouse devices automatically. The system uses three sensors—DHT11 temperature and humidity sensor, soil moisture sensor, and BH1750 light intensity sensor—to maintain optimal growing conditions for plants.

At the beginning of the process, all sensors are initialized, and the ESP32 microcontroller starts collecting data from each sensor. The ESP32 acts as the central processing unit, receiving inputs, comparing them with predefined threshold values, and controlling the corresponding output devices.

First, the DHT11 sensor measures the temperature and humidity inside the greenhouse. The sensor data is sent to the ESP32, where it is compared with the preset temperature limit (for example, 30°C). If the temperature is greater than the threshold value, the condition is evaluated as NO, and the ESP32 turns ON the exhaust fan. This helps in removing excess heat and humidity, thereby maintaining a suitable environment. If the temperature is within the acceptable range, the condition is evaluated as YES, and the exhaust fan remains OFF.

Next, the soil moisture sensor measures the moisture content of the soil and sends the analog value to the ESP32. This value is compared with a predefined soil moisture threshold (for example, 2500). If the soil moisture value is below the threshold, indicating dry soil, the condition becomes YES, and the ESP32 activates the water pump to irrigate the plants. If sufficient moisture is detected, the condition becomes NO, and the pump remains OFF, preventing overwatering.

Finally, the BH1750 light intensity sensor measures the ambient light level inside the greenhouse in terms of lux. The ESP32 compares this value with the required light threshold (for example, 300 lux). If the light intensity is less than the required value, the condition is evaluated as YES, and the ESP32 turns ON the LED grow light to provide adequate illumination for photosynthesis. If sufficient light is available, the condition is evaluated as NO, and the LED remains OFF, conserving energy.

This process runs continuously in a loop, allowing real-time monitoring and automatic adjustment of greenhouse conditions. The flowchart clearly represents the decision-making process of the system, where each sensor input leads to a specific control action. By following this flow, the smart greenhouse system ensures efficient climate control, proper irrigation, and adequate lighting with minimal human intervention.

Algorithm

- **Start**
- Initialize the ESP32 microcontroller.
- Initialize the DHT11 temperature and humidity sensor.
- Initialize the soil moisture sensor.
- Initialize the BH1750 light intensity sensor.
- Set threshold values for temperature, soil moisture, and light intensity.
- Read temperature and humidity values from the DHT11 sensor.
- If temperature is greater than the threshold value (e.g., 30°C),
    - Turn **ON** the exhaust fan.
      Else,
    - Turn **OFF** the exhaust fan.
- Read soil moisture value from the soil moisture sensor.
- If soil moisture value is less than the threshold (e.g., 2500),
    - Turn **ON** the water pump.
      Else,
    - Turn **OFF** the water pump.
- Read light intensity value from the BH1750 sensor.
- If light intensity is less than the threshold (e.g., 300 lux),
    - Turn **ON** the LED light.
      Else,
    - Turn **OFF** the LED light.
- Repeat steps 7 to 12 continuously.
- **Stop**.

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Working Methodology

The working methodology of the Smart Greenhouse Automated System is based on continuous monitoring of environmental parameters and automatic control of greenhouse devices using an ESP32 microcontroller. The system follows a structured flow as shown in the flowchart, ensuring real-time response to changing greenhouse conditions.

Initially, the system is powered ON and the ESP32 microcontroller initializes all connected sensors, namely the DHT11 temperature and humidity sensor, soil moisture sensor, and BH1750 light intensity sensor. Predefined threshold values for temperature, soil moisture, and light intensity are stored in the ESP32 program, which represent the optimal environmental conditions required for plant growth.

Once initialization is complete, the ESP32 begins reading temperature and humidity data from the DHT11 sensor. The measured temperature value is compared with the predefined threshold (for example, 30°C). If the temperature exceeds the threshold, the ESP32 identifies this as an unfavorable condition and activates the exhaust fan through a relay module. The exhaust fan helps in reducing excess heat and humidity by improving air circulation inside the greenhouse. If the temperature remains within the acceptable range, the exhaust fan remains switched OFF.

Next, the ESP32 reads the soil moisture value from the soil moisture sensor. The sensor provides an analog value corresponding to the water content in the soil. This value is compared with the preset soil moisture threshold. If the soil moisture level is below the threshold, indicating dry soil conditions, the ESP32 turns ON the water pump to irrigate the plants. Once sufficient moisture is detected, the pump is automatically turned OFF, preventing overwatering.

Following irrigation control, the ESP32 reads light intensity data from the BH1750 sensor using I^2C communication. The light intensity value is compared with the required minimum light level for plant growth. If the light intensity is found to be insufficient, the ESP32 switches ON the LED grow light to provide additional illumination. When adequate light is available, the LED is turned OFF to conserve energy.

This entire process operates continuously in a loop, allowing the system to respond instantly to environmental changes. By following this methodology, the smart greenhouse system ensures optimal temperature, proper irrigation, and adequate lighting with minimal human intervention. The flowchart-driven methodology provides a simple, efficient, and reliable automation approach for modern greenhouse management.

Result

The Smart Greenhouse Automated System using ESP32 was successfully designed, implemented, and tested to monitor and control essential environmental parameters required for plant growth. The system effectively integrated the DHT11 temperature and humidity sensor, soil moisture sensor, and BH1750 light intensity sensor to automate greenhouse operations with minimal human intervention.

During testing, the **DHT11 sensor** accurately monitored the temperature and humidity inside the greenhouse environment. When the temperature or humidity exceeded the predefined threshold values, the ESP32 responded promptly by activating the **exhaust fan** through the relay module. The fan operation helped in reducing excess heat and moisture, thereby maintaining a stable and suitable internal climate. When the temperature and humidity returned to normal levels, the exhaust fan was automatically switched OFF, demonstrating reliable and responsive environmental control.

The **soil moisture sensor** continuously measured the water content in the soil. When the soil moisture level dropped below the set threshold, the ESP32 automatically turned ON the **water pump**, supplying water to the plants. Once the soil moisture reached the required level, the pump was turned OFF automatically. This ensured efficient irrigation, prevented overwatering and underwatering, and significantly reduced water wastage.

The **BH1750 light intensity sensor** provided accurate real-time measurement of ambient light intensity in lux. When insufficient light conditions were detected, the ESP32 successfully activated the **LED grow light** to provide adequate illumination for plant growth. The LED was turned OFF automatically when sufficient light was available, ensuring optimal energy usage and consistent lighting conditions.

The system operated continuously and reliably under varying environmental conditions. The relay module effectively isolated high-power devices such as the exhaust fan and water pump, ensuring safe and stable operation. Overall, the system demonstrated efficient automation, precise sensor-based decision-making, and effective control of greenhouse parameters.

The results confirm that the Smart Greenhouse Automated System successfully maintains optimal environmental conditions, improves resource utilization, and reduces the need for manual supervision. The project proves the feasibility and effectiveness of using ESP32-based automation for modern smart agriculture applications.

Future Scope

1. **IoT-Based Remote Monitoring**
   Integration with cloud platforms will allow farmers to monitor and control greenhouse conditions remotely using mobile or web applications.
2. **Artificial Intelligence and Machine Learning**
   AI algorithms can predict plant needs and automatically adjust temperature, irrigation, and lighting for maximum crop yield.
3. **Advanced Sensor Integration**
   Future systems can include CO_2, pH, nutrient, and gas sensors for more precise environmental control.
4. **Computer Vision and Image Processing**
   Cameras and image analysis can be used to detect plant diseases, pests, and growth stages automatically.
5. **Renewable Energy Usage**
   Solar and wind energy can be integrated to reduce power consumption and make greenhouses energy efficient.
6. **Autonomous Robotics**
   Robots can be used for planting, pruning, harvesting, and greenhouse maintenance, reducing labor dependency.
7. **Data Analytics and Crop Forecasting**
   Big data analytics can help in forecasting crop yield and optimizing greenhouse performance.
8. **Automated Nutrient Management**
   Smart fertigation systems can automatically supply nutrients based on plant growth stages and soil conditions.

9. **Scalable and Modular Design**
   Future systems can be easily expanded for large-scale commercial greenhouses with minimal modification.
10. **Digital Twin Technology**
   Virtual greenhouse models can simulate environmental conditions and improve decision-making before real-world implementation.

Conclusion

The Smart Greenhouse Automated System using ESP32 was successfully designed and implemented to monitor and control essential environmental parameters required for efficient plant growth. The system effectively integrates the DHT11 temperature and humidity sensor, soil moisture sensor, and BH1750 light intensity sensor to automate greenhouse operations. Based on real-time sensor data, the ESP32 controls an exhaust fan to regulate temperature and humidity, a water pump to ensure proper irrigation, and an LED light to maintain adequate lighting conditions inside the greenhouse.

The automated operation of the system reduces the need for continuous manual monitoring and minimizes human error. Efficient control of irrigation prevents water wastage, while automated ventilation and lighting help maintain a stable and suitable environment for plant development. The use of relay modules ensures safe control of high-power devices, making the system reliable and user-friendly.

Overall, the project demonstrates the effective application of embedded systems and sensor-based automation in modern agriculture. The Smart Greenhouse Automated System provides a cost-effective, efficient, and scalable solution for greenhouse management and serves as a strong foundation for future enhancements such as IoT-based remote monitoring and intelligent control systems.